

# Gangmin Yu

Mechanical Engineering · Undergraduate Researcher

Kyonggi University · RVLab

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## RESEARCH INTERESTS

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Autonomous driving and navigation, vehicle dynamics and control, radar/sensor perception, robot simulation and digital twins, embedded and mechatronic systems.

## EDUCATION

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**Kyonggi University**, Suwon, Republic of Korea

Mar. 2021 – Present

B.S. in Mechanical Engineering

Undergraduate Researcher, RVLab (Advisor: Prof. Minseong Choi)

**GPA:** 4.27 / 4.5   **Major GPA:** 4.41 / 4.5

**English:** TOEIC Speaking IH (150)

## PUBLICATIONS

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### Domestic Conferences

이충헌, 유강민, 박영상, 최민성, 조우성. “Swerve 구동 휴머노이드를 위한 Holonomic 실내 자율 탐색 알고리즘” 제어·로봇·시스템학회(ICROS) 학술대회, 2026.

김도윤, 강민수, 유강민, 서형태, 박영상, 최민성. “IsaacSim 기반 휴머노이드 ACT 학습제어를 위한 VR 원격조작 및 Jitter 저감 기법” 제어·로봇·시스템학회(ICROS) 학술대회, 2026.

성민준, 유강민, 김준규, 한승호, 양승훈, 최민성. “도로 노면 의미론적 분할 기법의 소수 클래스 불균형 해소를 위한 원근 변환 기반 데이터 증강 기법” 제어·로봇·시스템학회(ICROS) 학술대회, 2026.

## RESEARCH & PROJECTS

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### ONGOING

#### 4D Radar Ego-Velocity Estimation

RVLab, Kyonggi University

- Estimating ego-vehicle velocity directly from 4D radar measurements toward robust self-motion estimation for autonomous driving.

#### Manipulation & Navigation for Humanoids in Domestic Environments

RVLab

- Applied physics-based digital-twin simulation and MetaQuest 3 teleoperation for a door-opening task in Isaac Sim.
- Implemented autonomous driving and navigation using AMCL localization on a generated 2D LiDAR map.

### PAST

#### 3D Simulation Environments & Physics Modeling for Daily-Life Scenes

KIMM (commissioned)

funded by Korea Institute of Machinery & Materials (KIMM)

- Built physically simulated indoor environments in NVIDIA Isaac Sim (PhysX) for realistic robot interaction.
- Authored rigid-body and collider models from .obj meshes; set per-object mass, inertia, friction, restitution.
- Defined articulated structures (hinge/prismatic joints) for doors and drawers with motion limits and dynamics.

### Autonomous Vehicle Build

*Autonomous Driving Club*

- Led the overall build of an autonomous vehicle; designed the hardware/embedded system (actuators, sensors, MCU communication).
- Implemented steering control (Pure Pursuit) and speed control (PID).

### High-Voltage Shutdown Circuit

*Self-Built Car Club · Formula (EV)*

- Designed a safety shutdown circuit for 120 V high voltage, cutting power on BMS/IMD/interlock faults or sudden-acceleration detection.
- Designed and fabricated the PCB (KiCad) and applied it to the vehicle.

### Braking System Design

*Self-Built Car Club · Formula & Baja*

- Computed braking force from vehicle kinematics and designed the brake disc accordingly.
- Used CAE to optimize the design and verify structural safety.

## AWARDS

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**Excellence Award** — Self-Built Autonomous Vehicle Division

*Nov. 2025*

University Creative Mobility Competition · Korea Transportation Safety Authority (KOTSA)

**Acceleration Award**

*Nov. 2021*

International Self-Built Car Competition · Yeungnam University

## TECHNICAL SKILLS

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**Simulation / Robotics:** NVIDIA Isaac Sim, PhysX, ROS2, AMCL / Navigation

**Control / Embedded:** Embedded C, PID, Pure Pursuit, MCU & Sensors

**Design / CAE:** KiCad (Schematic / PCB), AutoCAD, CATIA, Ansys

**Domain:** Autonomous Driving, Vehicle Dynamics, Radar Perception, Mechatronics